A Mini Project Report of

PCC-201-COM : Data Structures

Submitted by

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CERTIFICATE

This is to certify that the **"Mini Project Report"** submitted by **VITTHAL ANANT POTDAR (61)**, on **"Implementation of Snake and Ladder [BFS]"** is work done by him and submitted during **2026–2027** academic year, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in Computer Engineering, at **Bharati Vidyapeeth's College of Engineering, Lavale, Pune.**

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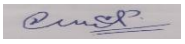
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Student Name



ABSTRACT

The Snake and Ladder game is a popular board game in which players move from a starting position to a destination by rolling a dice. Snakes move players backward, whereas ladders help players move forward.

This project implements the Snake and Ladder game using the Breadth-First Search (BFS) algorithm in Python. The board is represented as a graph, where each square represents a vertex and each possible dice movement represents an edge.

BFS explores the graph level by level and determines the minimum number of dice throws required to reach the final square. The project demonstrates the application of graph traversal algorithms in solving shortest-path problems.

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1. INTRODUCTION

1.1 Overview:

The project “Snake and Ladder using BFS” is designed to calculate the least number of dice rolls required to reach the last square on the board. The board is modeled as a graph, with each square representing a node and every possible dice roll representing a connection between nodes. When a player lands on a snake or ladder, their position changes according to its location. BFS explores the possible moves level by level to determine the shortest route to the destination.

1.2 Problem Statement:

The aim of this project is to develop a Python program that calculates the minimum number of dice throws required to reach the last square in a Snake and Ladder game. The program considers all possible dice outcomes along with the movements caused by snakes and ladders. By applying the Breadth-First Search (BFS) algorithm, the system explores possible positions systematically and identifies the most efficient way to reach the destination.

1.3 Objective:

The objective of this project is to implement the Snake and Ladder game as a graph -search problem using Breadth-First Search in Python. The project aims to:

- Represent each board position as a graph vertex.
- Model dice rolls as possible transitions between positions.
- Apply snake and ladder mappings whenever a move reaches their starting cells.
- Use a queue and visited set to find and reconstruct a shortest path with the minimum dice throws.

2. LITERATURE SURVEY

Graph algorithms are widely applied in solving problems that involve movement between different positions and finding efficient routes. Breadth-First Search (BFS) is a graph traversal technique that visits nodes in successive levels and is effective for finding the shortest path in an unweighted graph. This literature survey presents research related to graph traversal methods, pathfinding techniques, and their use in solving the Snake and Ladder problem.

Table 2.1 summarizes the main references that form the theoretical background of this project.

Year	Paper Name	Author	Conclusion
2025	Research on Strategy Optimization for the Snakes and Ladders Game Based on Graph Theory and Markov Chains	Yaowei Zhang	The research examines how graph theory can represent the Snake and Ladder board and help identify efficient routes. It also considers Markov chains for analyzing game strategies.
2025	A Review of Graph Traversal Algorithms: Techniques and Applications in Network Analysis	Chreesk Sabah M. Ali and Ibrahim Mahmood Ibrahim	The paper explains BFS and DFS techniques, highlighting their working methods, efficiency, strengths, and limitations in network analysis..
2025	Comparative Analysis between BFS and DFS-Shortest Path Algorithms	Renad AlKahtani et al.	The research studies the differences between BFS and DFS in pathfinding and evaluates their suitability for different graph-based problems.
2025	Simulation-based Review of Classical, Heuristic, and Metaheuristic Path Planning Algorithms	Kenneth Christopher Ugwoke, Nwojo Agwu Nnanna, and Saleh El-Yakub Abdullahi	The study discusses various path-planning methods and explains how they can be used to find routes between source and destination points.

3. UML DIAGRAM

The UML class diagram below represents the Snake and Ladder BFS solver, including the snake and ladder mappings, BFS search procedure, queue, visited set and parent map used to reconstruct the shortest path.

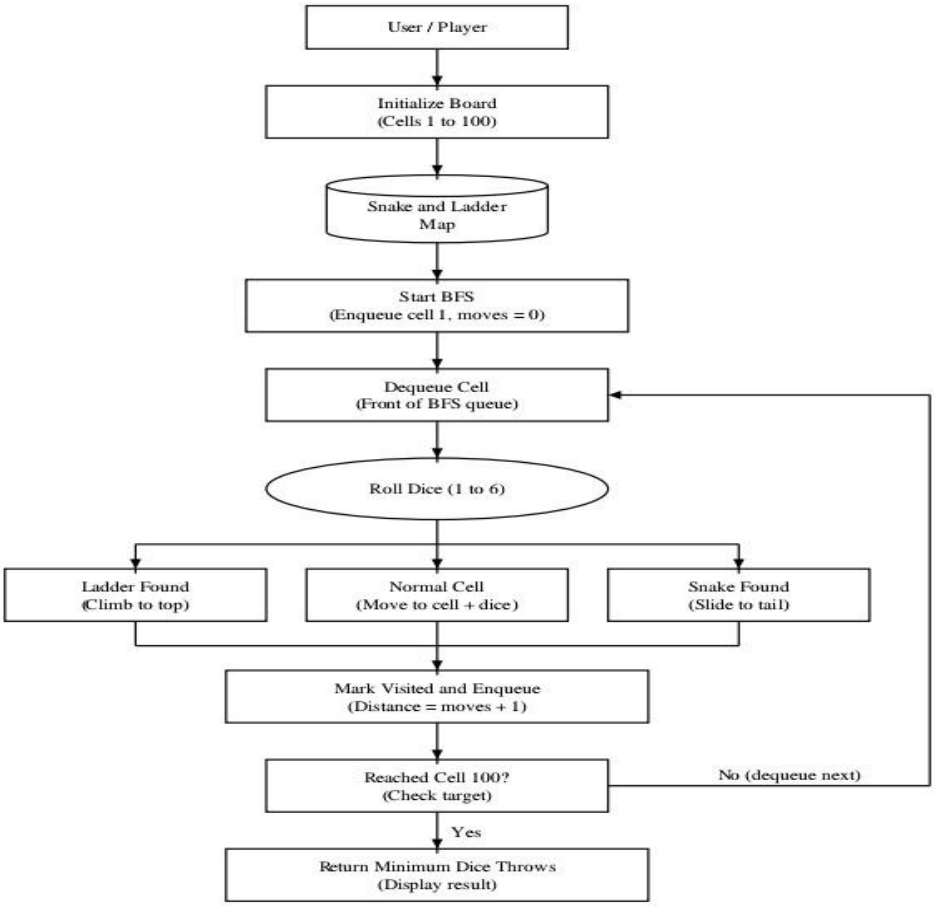


Fig 3.1: UML Class Diagram

4. SYSTEM REQUIREMENT

4.1 HARDWARE REQUIREMENTS

1. Operating System: Windows 11 64-bit
2. Processor Type: Intel Core
3. Recommended: Core 2 Duo or higher
4. Processor speed: 2.30 GHz or higher
5. RAM: Minimum 512 MB
6. Hard disk: 512 GB or more

4.2 SOFTWARE REQUIREMENTS

1. Jupyter Notebook (JupyterLab)
2. Language: Python 3 (ipykernel)
3. Python modules: re, collections

5. SYSTEM DESIGN

Existing System:

In a basic approach, the words of a text are stored in a list and, for every word, the list is searched to find whether the word already exists so that its count can be increased. This linear search needs O(n) time for every word, so counting a whole document takes $O(n^2)$ time in the worst case. Manual counting or spreadsheet formulas are also slow and error-prone for large texts, and they do not remove common words automatically.

Key approaches commonly used for solving the Snake and Ladder path problem:
Manual play (depends on repeated dice rolls)

- ✓ Brute-force sequence exploration (may repeat states)
- ✓ Breadth-First Search using a queue (explores states level by level)
- ✓ Visited set and parent map (avoid revisits and reconstruct the path)
- ✓ Library classes such as collections.Counter

Proposed System:

The proposed system models the board as a graph. Starting from cell 1, BFS examines each possible dice result from 1 to 6, applies snake or ladder transfers, and adds each unvisited destination to the queue. The parent map is used to reconstruct the shortest route once the final cell is reached.

- ✓ Initialize the queue with cell 1 and mark it visited.
- ✓ Generate valid dice-roll destinations for the current cell.
- ✓ Apply a snake or ladder transfer to each destination.

Time Complexity:

Operation	Time	Space
Tokenize text (n words)	O(n)	O(n)
Count with hash table	O(n) average	O(k)
Sort k unique words	O(k log k)	O(k)
top_n (count + sort + slice)	O(n + k log k)	O(k)

Working Diagram:

Fig 5.1 shows the working of the system. BFS starts at cell 1, generates possible dice moves, applies snake or ladder transfers, marks new cells as visited, and continues with a queue until the final cell is found. The parent map then reconstructs the shortest path.

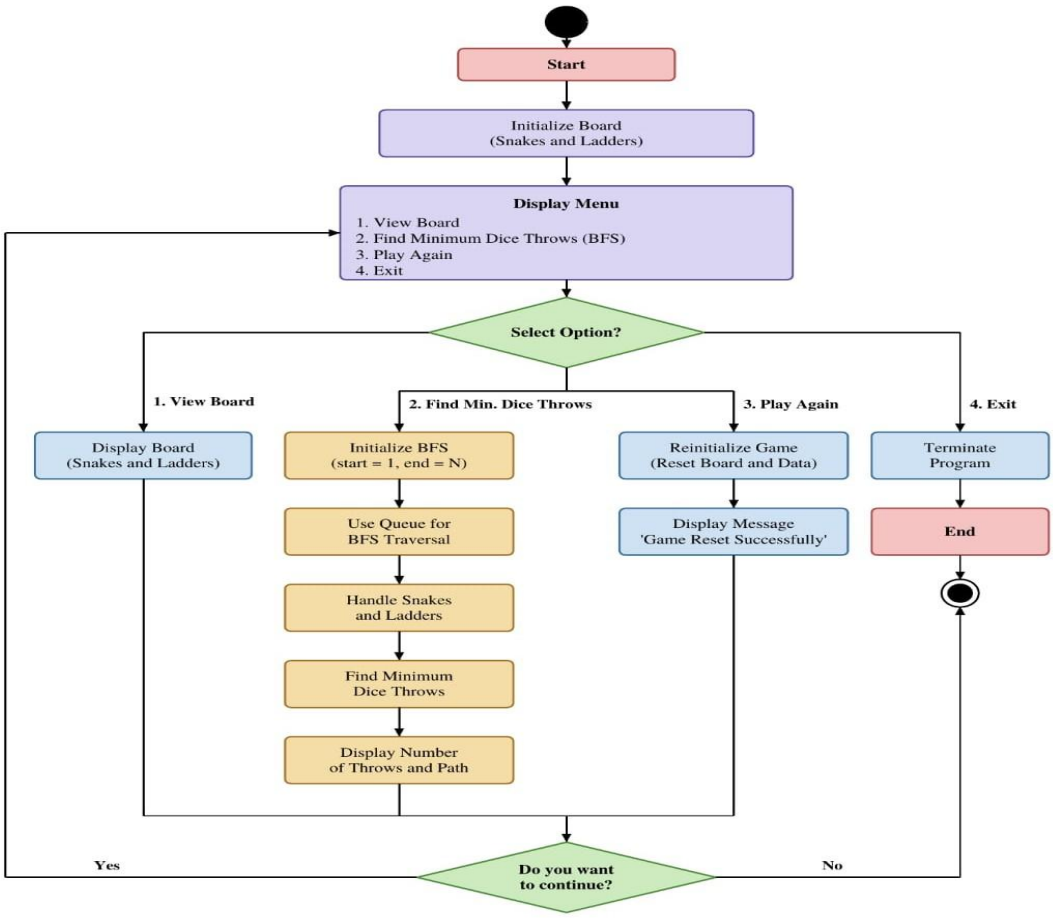
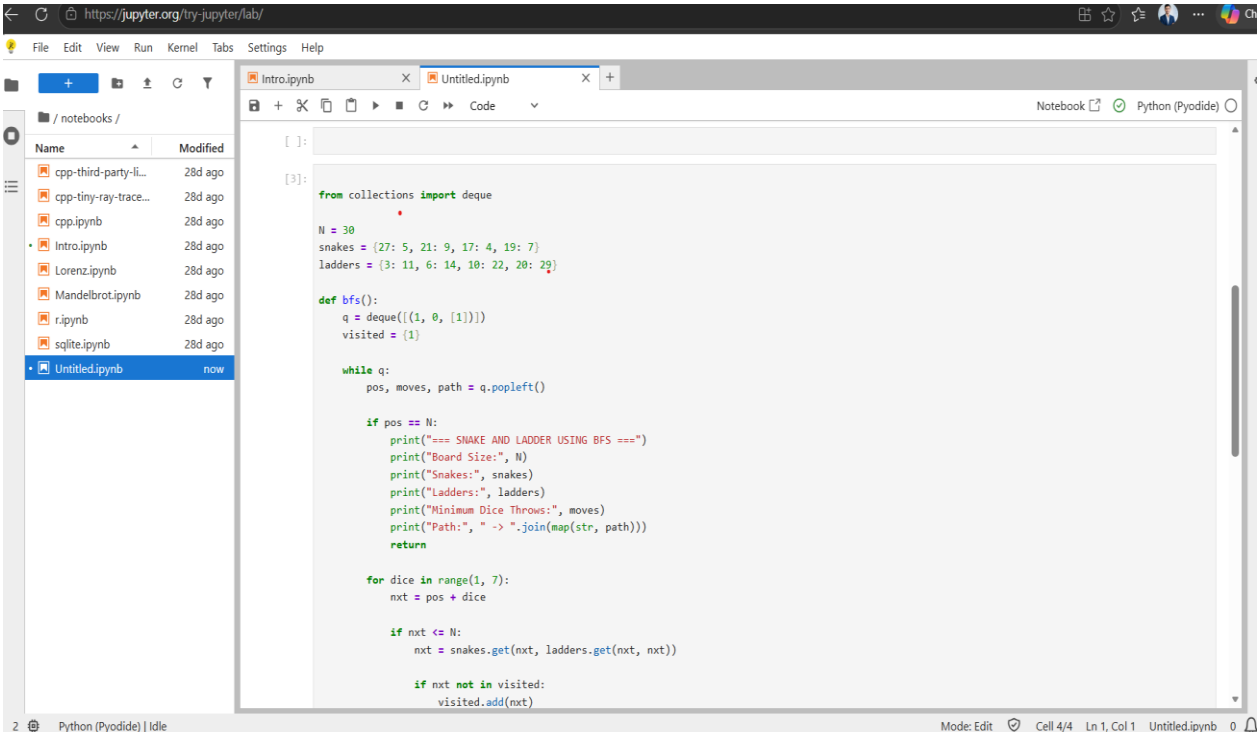


Fig 5.1: Working Diagram of Snake and Ladder BFS

SYSTEM CODE



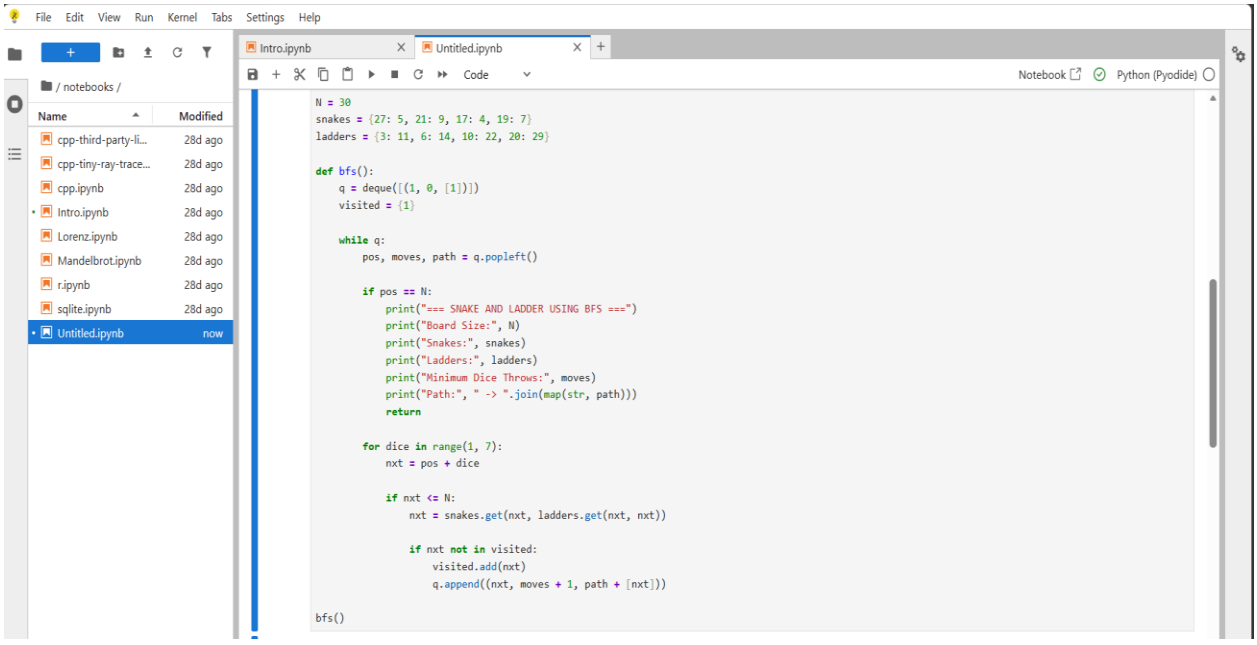


Fig 5.2(a): System code – snake/ladder mappings and BFS function

6. RESULT

The program explores reachable board positions using BFS and reports the minimum number of dice throws required to reach the final square. Each position is visited at most once, and the stored parent information is used to display the path found. The result depends on the snake and ladder positions defined for the board.

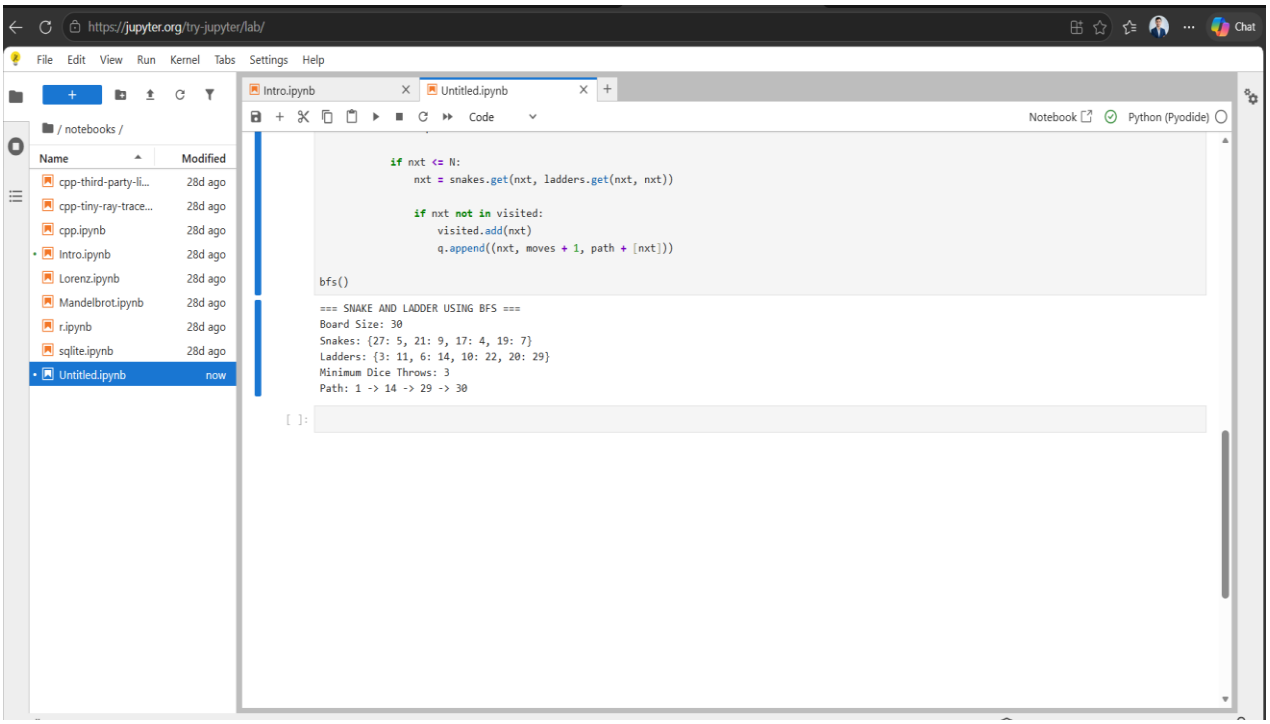


Fig 6.1: Result (Minimum Dice Throws and Shortest Path)

7. APPLICATIONS

Breadth-First Search and graph modelling can be applied to problems that require finding a shortest path in an unweighted graph. Typical applications include:

- Finding shortest paths in unweighted networks and maps.
- Solving puzzles and board games by representing states as graph vertices.
- Analyzing customer reviews, feedback and social media posts to find frequently discussed topics.
- Plagiarism detection and document similarity comparison.
- Exploring connected states in simulations and simple game AI.
- Log analysis and counting the occurrence of events, errors or user actions.

8. FUTURE WORK

The project can be improved and expanded in the following ways:

- Develop a graphical Snake and Ladder board to display snakes, ladders, and the player's movement.
- Provide options to customize the board size and set different snake and ladder positions.
- Compare BFS with other search algorithms to analyze their efficiency and performance.
- Create an interactive interface with dice rolling, player movement, and visualization of the shortest path.
- Extend the program to support multiple players, display game statistics, and record the minimum number of dice throws for different board configurations.

9. CONCLUSION

In this project, the Snake and Ladder game was implemented using the Breadth-First Search (BFS) algorithm. The board was modeled as a graph, where each square represents a node and each possible dice roll represents a movement between nodes. A queue was used to explore possible positions, while a visited set helped prevent repeated exploration.

The project demonstrates that BFS can determine the minimum number of dice throws required to reach the final square. It also provides practical knowledge of graph representation, queue operations, and shortest-path algorithms. Overall, this project helps in understanding how graph traversal techniques can be applied to solve real-world problems efficiently.

10. REFERENCES

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